



UNIVERSITY OF PATRAS
COMPUTER ENGINEERING AND INFORMATICS DEPARTMENT
MSc PROGRAM *DATA DRIVEN COMPUTING AND DECISION MAKING*
MSc topics 2022-2023
Georgios Drakopoulos

1. Computational infrastrure for temporal neural graph networks

Numerous social media problems can be naturally be represented by graphs. Because of the volume and structure of social media data, they must be processed by the appropriate graph engineering algorithms running on suitable computational infrastructure. Temporal graph neural networks (TGNNs) by design can mine interconnected data and extract knowledge through iteratively updating ground truth vectors leading to the emergence of a global steady state through a sequence of ephemeral topologies. The primary research objective is the deployment of a TGNN on a cluster of physical computers managed by Kubernetes, the latter allowing the use of off the shelf hardware. The application will be the vertex classification of a real world graph.

2. Data mining of music data

The advent of digital technologies has resulted in data abundance across various domains including the music industry allowing therefore for obtaining insight into consumer preferences and pop culture trends. By systematically mining music datasets and leveraging machine learning models, recommendations can be made to the industry. The primary research objective is the development of a stochastic variant of k-means algorithm to handle efficiently large datasets which cannot be loaded in the main memory at once. As a secondary objective, guidelines for the Python implementation will be provided. The evaluation metrics will be scalability and computational efficiency.

3. SkyTrack: From simulations to tracking algorithm testing

AI applications range from recommendations to complex autonomous driving systems. The rapid evolution in computing power and efficiency has allowed the use of neural networks models on mobile devices and generally in low energy consumption devices. This MSc thesis presents a comprehensive target detection and monitoring system using an unmanned aerial vehicle (UAV). This system uses the YOLO-V4 object detection model and the DeepSORT identification correlation code. This system has been named SkyTrack and is validated in simulations and in the real world. Initially, the simulation environment were ROS, GAZEBO and an Iris vehicle are used with some modifications is presented. In these tests there will be also obstacle avoidance logic by using the ORCA algorithm. Then tests will be done using the DJI Phantom 3 and MINI 2 UAV.

Note: This topic was proposed by the student.